Geometric Group Theory

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Lecture 1

1 Combinatorial group theory

Combinatorial group theory is a sibling field to Geometric group theory. Both study infinite discrete groups.

1.1 Free groups and presentations

Let $A = \{a_1, a_2, \dots\}$ be an *alphabet*. A group F is *free on* A if

1. There is a map of sets $A \to F$,
2. for any group G, and a map of sets $A \to G$, there exists a unique group homomorphism $F \to G$ such that

$$\begin{array}{ccc} & & F \\ & \nearrow & \downarrow \\ A & \longrightarrow & G \end{array}$$

commutes.

This is a *universal property*. As usual, this means that F is unique up to unique isomorphism. This shows that F is determined by A, so we may write $F = F(A)$.

However (as usual with definitions by universal property), we don't know if $F(A)$ exists. We'll show this two different ways.

1. **Topologically:** Let

$$X = \bigvee_{a \in A} S^1$$

By the Seifert-van Kempen theorem,

$$\pi_1(X) \cong F(A)$$

2. **Combinatorially:** Let

$$A^* = \{\text{words in } A \sqcup A^{-1}\}$$

where $A^{-1} = \{a_1^{-1}, \dots, a_n^{-1}\}$. For example,

$$1 = \varnothing, aa, aa^{-1}, aba^{-1}b^{-1}, a^{100}ba^{-100}b, \dots$$

A word is *reducible* if it contains aa^{-1} or $a^{-1}a$ as a subword for any $a \in A$. Otherwise, it is *reduced*. We can now define

$$F(A) = \{w \in A^* \text{ reduced}\}$$

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The group operation is concatenation, followed by reduction. For example,

$$(abab^{-1})(b^2a) = abab^{-1}b^2a = ababa$$

Note that reduction terminates as each reduction step reduces the length. We won't check that this is well defined or associative. The identity element is 1, inverses is clear.

A *presentation* consists of an alphabet A, which we will call *generators*, and a set $R \subseteq F(A)$, which we will call *relations*, and we write

$$\langle A \mid R \rangle = \langle a_1, a_2, \cdots \mid r_1, r_2, \ldots \rangle$$

This *presents* a group

$$G = \frac{F(A)}{\langle\langle R \rangle\rangle}$$

where $\langle\langle R \rangle\rangle$ is the *normal closure* of R, i.e. the smallest normal subrgoup of $F(A)$ containing R.

Example 1.1

Some examples of presentations which we have seen:

$$\begin{aligned}
\langle a \mid a^n \rangle &\cong \mathbb{Z}/n\mathbb{Z} \\
\langle r, s \mid r^n, s^2, srsr \rangle &\cong D_{2n} \\
\langle A \mid \rangle &\cong F(A) \\
\langle a, b \mid aba^{-1}b^{-1} \rangle &\cong \mathbb{Z}^2 \\
\langle a_1, \ldots, a_g, b_1, \ldots, b_g \mid a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1} \rangle &\cong \pi_1(\Sigma_g)
\end{aligned}$$

where Σ_g is the compact orientable surface of genus g.

As we see, presentations arise when we write down fundamental groups of spaces. In fact, all groups arise this way.

Corollary 1.2 (of Seifert-van Kampen)**.** For

$$G = \langle a_1, a_2, \cdots \mid r_1, r_2, \ldots \rangle$$

there exists a space X with $\pi_1(X) = G$.

Proof. First, start of with a wedge of circles, one for each a_i. Also consider a disjoint union of discs, one for each r_i. Attach the i-cell along its boundary, which is a loop in the wedge of circles. □

This is called the *presentation complex* of G. For example, if we have $\mathbb{Z}^2 = \langle a, b \mid aba^{-1}b^{-1} \rangle$, then $X = \mathrm{T}^2$ is the 2-torus.

In 1911, Max Dehn posed the following problems.

1. (Word problem) Given $w \in A^*$, determine whether or not $w = 1$ in $G = \langle A \mid R \rangle$. That is, whether or not $w \in \langle\langle R \rangle\rangle \subseteq F(A)$.
2. (Conjugacy problem) Given $G = \langle A \mid R \rangle$, $u, v \in A^*$, determine whether or not u is conjugate to v in G.
3. (Isomorphism problem) Given $G = \langle A \mid R \rangle$, $H = \langle B \mid S \rangle$, determine whether or not G is isomorphic to H.

Lecture 2

Remark 1.3. The conjugacy problem is stronger than the word problem, since $w = 1$ if and only if w is conjugate to 1.

Dehn was motivated by topology, but the problems asks for algorithms. We will often solve them using geometry.

All three were *un*solved in the 1950s, as all three problems are algorithmically undecidable. Nevertheless, positive solutions are known for many reasonable classes of groups.

Let $A = \{a_1, \ldots, a_n\}$ be a finite alphabet.

Example 1.4 (word problem in free groups)

Let $w \in A^*$. If w is reduced, then $w = 1$ if and only if $w = \varnothing$. Otherwise, w contains a subword aa^{-1} for some $a \in A \cup A^{-1}$. Cancelling aa^{-1} produces a word $w' \in A^*$, such that $w = w'$ in $F(A)$, and $\ell(w') = \ell(w) - 2 < \ell(w)$. This terminates after finitely many steps.

We can also solve the conjugacy problem for free groups.

Definition 1.5

There is a natural action of $\mathbb{Z}$ on A^* permuting words. That is,

$$1 \cdot a_1 \cdots a_k = a_2 a_3 \cdots a_k a_1$$

The elements of $\mathbb{Z} \cdot w$ are called the *cyclic conjugates* of w. Note that all $u \in \mathbb{Z}w$ are conjugate to w. The orbits $\mathbb{Z}\backslash A^*$ are called *cyclic words*. A word is *cyclically reduced* if every cyclic conjugate is reduced.

Example 1.6

aba^{-1} is reduced, but not cyclically reduced as $ba^{-1}a$ is not reduced.

Note that if w is reduced, but not cyclically reduced, then

$$w = aw'a^{-1}$$

for some $a \in A \cup A^{-1}$. Note that w' is both conjugate to w and shorter than w. Hence after finitely many iterations, we can assume that w is cyclically reduced.

Lemma 1.7 (conjugacy problem in free groups)**.** If $u, v \in F(A)$ are cyclically reduced, then u is conjugate to v if and only if the corresponding cyclic words are the same.

Proof. If u, v have the same cyclic words, then v is a cyclic conjugate of u, which we have seen is a conjugate of u.

Conversely, suppose $u = gvg^{-1}$. By induction on $\ell(g)$, we can assume $g = a \in A \cup A^{-1}$. From this it follows that either $v = a^{-1}v'$, or $v = v'a$, as v is cyclically reduced. That is, $u = v'a^{-1}$ or $u = av'$. In both cases, they are cyclic conjugates. □

1.2 Historical case study

Let's briefly think about the state of topology in the early 20th century. Poincaré knew that homology classifies the compact two-dimensional surfaces. This motivated the

Conjecture 1.8 (Poincaré conjecture, version 1)**.** Let M be a compact connected 3-manifold, with

$$\mathrm{H}_*(M) = \begin{cases} \mathbb{Z} & * = 0, 3 \\ 0 & \text{otherwise} \end{cases}$$

Then M is homeomorphic to S^3.

Such a 3-manifold M is called a *homology sphere*.

Theorem 1.9 (Poincaré, 1904)**.** There exists a three dimensional homology sphere P, with

$$\pi_1(P) \twoheadrightarrow A_5$$

The moral is that: homology is not enough, we need use π_1 as well.

Conjecture 1.10 (Poincaré conjecture, version 2)**.** Let M be a compact connected 3-manifold, with $\pi_1(M) = 1$. Then M is homeomorphic to S^3.

This was proven by Perelman in 2003. Returning to the original conjecture, in 1910 Dehn wanted to construct more homology spheres.

Theorem 1.11 (Dehn, 1910)**.** There are infinitely many non-homeomorphic 3-dimensional homology spheres.

Remark 1.12. The isomorphism problem is exactly what is needed to distinguish these manifolds.

Lecture 3

Here is Dehn's construction. Consider the trefoil knot $K \subseteq \mathbb{R}^3 \subseteq S^3$. Let $N(K)$ be a regular open neighbourhood of K, i.e. a thickening of K. Let $N = S^3 \setminus N(K)$. This is a compact 3-manifold with boundart $\partial N \cong \mathrm{T}^2$.

Computing,

$$\mathrm{H}_*(N) = \begin{cases} \mathbb{Z} & * = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\pi_1(N) \cong \langle x, y, z \mid x^2 = y^3 = z \rangle$$

This follows from the fact that K is a torus knot, and we can compute the π_1 of the complement of a torus knot. Note that the abelianisation map

$$\pi_1(N) \to \mathrm{H}_1(N) \cong \mathbb{Z}$$

sends z to 6, x to 3 and y to 2. Moreover, the boundary torus T has $\pi_1(T) = \mathbb{Z}^2$, generated by xy and z. Under the abelianisation map, xy is sent to 5.

We can glue a solid torus $D^2 \times S^1$ to N, by a homeomorphism on the boundary. Let λ, μ be the corresponding loops on T^2. The resulting manifold $M_\phi = N \cup_\phi T$ is closed. $\pi_1(M_\phi)$ depends on $g = \phi_*(\mu)$. By Seifert-van Kapmen,

$$\pi_1(M_\phi) = \frac{\pi_1(N)}{\langle\langle g \rangle\rangle}$$

Similarly,

$$\mathrm{H}_1(M_\phi) = \frac{\mathbb{Z}}{\langle [g] \in \mathrm{H}^1(N) \rangle}$$

To produce a homology sphere, we need to choose ϕ such that $g = \phi_*(\mu) = 1$ in $\mathrm{H}^1(N)$.

If $g = (xy)^a z^b$, then in $\mathrm{H}^1(N)$ this is mapped to $5a + 6b \in \mathbb{Z}$. Choose $a = 6n + 5$, and $b = -(5n + 4)$ for $n \in \mathbb{Z}$. He constructs ϕ_n such that

$$\phi_n(\mu) = g_n = (xy)^{6n+5} z^{-(5n+4)}$$

His family of manifolds

$$D_n = N \cup_{\phi_n} U$$

has

$$\pi_1(D_n) = \langle x, y, z \mid x^2 = y^2 = z, (xy)^{6n+5} = z^{5n+4} \rangle$$

The remaining challenge is to prove the groups $G_n = \pi_1(D_n)$ for $n \geq 0$ are pairwise non-isomorphic.

This is the isomorphism problem! In particular,

$$g_n = g_m \implies g_n \text{ and } g_m \text{ are conjugate} \implies G_n \cong G_m$$

So we also need to solve the word and conjugacy problem in $\pi_1(N)$.

1.3 van Kampen diagrams

Definition 1.13

A map $f : Y \to X$ of cell complexes is called *combinatorial* if for all $k \in \mathbb{Z}_{\geq 0}$, and every k-cell e^k of Y, f maps the interior $\mathrm{Int}(e^k)$ homeomorphically to the interior of a k-cell of X.

Consider a presentation $G = \langle a_i \mid r_j \rangle$, and the associated presentation complex X.

Definition 1.14

A *(singular) disc diagram* is a compact contractible 2-complex D, with an embedding $D \hookrightarrow \mathbb{R}^2$. A disc diagram D is *over* X if it is equipped with a combinatorial map $D \to X$.

Example 1.15

Example of disc diagram.

Recall that X is given by a wedge of circles, with discs glued on for each relation. So the 1-cells correspond to generators, 2-cells go to relations (or cyclic conjugates, or inverses). With this:

- every oriented 1-cell of D is labelled with some $a_i \in A$,
- so that each 2-cell has boundary which is a cyclic conjugate of some $r_j^{\pm 1}$.

Associated to each disc diagram D, we have a *boundary cycle*, which reads a (cyclic[1]) word $w \in A^*$, which reduces to an element $w' \in \langle\langle r_j \rangle\rangle \leq F(A)$. To see this, D is contractible.

D is a van Kampen diagram for w.

[1] or a word once we choose a base point.